A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Heavy metals: lead, mercury - Set 1

## Mark scheme

AQA spec 3.4.3.2.11; Pollution; textbook pp. 275-279

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** (15.0 - 2.2) / 15.0 x 100<br>**1** = 85.3% (accept 85%)<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (%) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Added as an anti-knock agent to smooth combustion and reduce engine wear<br>**1** Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food<br>**1** Lead is a neurotoxin that bioaccumulates<br>– chronic exposure damages the nervous system / brain development in children<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **particulates** → fine solid particles / smoke and dust | 3 |
| **01.3** | **Any 2 of 5**<br>1 mark each for any 2 of the 5 points below, in any order, maximum 2.<br>**1** Replacement of lead water pipes / lead solder with copper / tin-based solder<br>– phosphoric acid dosing of water supplies forming insoluble lead phosphate<br>**1** Phasing out of lead-based paint / safer paint removal<br>**1** Lead-free electrical solder<br>**1** Restrictions on lead fishing weights / shotgun pellets<br>**1** Industrial emission controls<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 2 |
| **01.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe<br>**1** This prevents lead dissolving into the water<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **01.5** | 1 mark for the point below.<br>**1** Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• quotas described as solving overfishing on their own - undersized fish are still caught and discarded | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Inorganic mercury settled into anaerobic sediments where anaerobic microbes converted it to organic (methyl) mercury<br>**1** Methyl mercury is liposoluble so it is absorbed easily and stored in fat / passes through cell membranes<br>**1** It bioaccumulated in fish and shellfish and biomagnified along the food chain to reach much higher concentrations than in the water<br>**1** Local people ate large amounts of fish (critical group) so received toxic doses<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **critical group** → the people most at risk / most heavily exposed<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 4 |
| **02.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Liposoluble so it crosses the blood-brain barrier into the brain, inhibiting enzymes / nerve function (neurotoxin)<br>**1** It crosses the placenta and harms the developing unborn baby<br>**1** Mercury is a teratogen, preventing normal gene expression / development<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 3 |
| **02.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Elemental (liquid) mercury: not easily absorbed through skin or gut, though vapour is absorbed in the lungs<br>**1** Inorganic compounds (eg mercury oxide): moderately well absorbed in the gut<br>**1** Organic compounds (eg methyl mercury): easily absorbed through skin, gut and lungs<br>– liposoluble so most toxic and bioaccumulates<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 276-277; AQA spec 3.4.3.2.11. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.1** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1 Disposal of items containing mercury:** batteries, thermometers, fluorescent lights<br>**1** Chemical plants producing chlorine using mercury electrodes<br>**1** Combustion of coal<br>**1** Gold mining (amalgamation)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 3 |
| **03.2** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Replacement of mercury thermometers with alcohol / electronic thermometers (discontinued uses)<br>**1** Removal from effluents by reverse osmosis / activated carbon filters / ion exchange filters<br>**1** Disposal / storage of wastes at high pH to reduce solubility<br>**1** Encapsulation of mercury-containing hazardous waste in cement<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 3 |
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Most heavy metals are more soluble at low pH / less soluble at high pH<br>**1** So at high pH they remain insoluble / immobile and cannot be leached into water or absorbed by organisms<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leached** → nutrients or ions washed down out of the soil / washed away by water<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** The effects of two pollutants interact to produce a different (greater) effect than either alone<br>**1** Cadmium and zinc<br>– ozone and sulfur dioxide<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any appropriate named example, including ones that are not listed here.<br>• **synergism** → two pollutants together doing more damage than the sum of each alone<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-278; AQA spec 3.4.3.2.11. | |
| **04.1** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** 180 / 1.2 = 150<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (times) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 1 |
| **04.2** | 1 mark for the point below.<br>**1** Cadmium and zinc occur together in the same ores so mine drainage / smelting releases both<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Mine drainage water contaminated irrigation water / paddy soil so rice absorbed cadmium (1.2 vs 0.05 mg $kg^{-1}$)<br>**1** Cadmium is liposoluble / not excreted so it bioaccumulates in the body (kidneys) with chronic exposure to small doses over years<br>**1** Cadmium causes kidney failure and bone decalcification / skeletal collapse (itai-itai disease)<br>**1** Rice is the staple food so residents receive a large dose<br>– village B has no exposure and no disease (correlation supports causation)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **correlation** → a relationship in which the two variables change together<br>• **causation** → one variable actually causing the change in the other<br>**Do not credit**<br>• a correlation presented as proof of cause - say that it suggests a link that further work would have to test<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 4 |
| **04.4** | **Any 2 of 4**<br>1 mark each for any 2 of the 4 points below, in any order, maximum 2.<br>**1** Collect mine drainage water and neutralise / raise pH with lime so cadmium precipitates and stays insoluble<br>**1** Cover / cap spoil heaps to prevent leaching and dust<br>**1** Recycle nickel-cadmium batteries / hazardous waste landfill<br>**1** Phytoremediation of contaminated paddy soils<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leaching** → nutrients or ions washed down out of the soil / washed away by water<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue<br>• 'landfill gas is just carbon dioxide' - it is roughly half methane, a far stronger greenhouse gas | 2 |
| **04.5** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Manufacture may release cadmium in effluents / dust<br>**1** Panels have a limited life<br>– disposal of old panels (landfill, incineration of pigments) could release cadmium unless recycled<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **incineration** → burning waste, usually with energy recovery<br>**Do not credit**<br>• 'solar panels do not work in cold countries' - output depends on insolation, and cells are slightly more efficient when cool<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11. | |
| **05.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** B | 1 |
| **05.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** TBT is highly toxic to marine molluscs / crustaceans and an endocrine disruptor altering growth and reproduction of oysters / whelks even at very low concentrations<br>**1** It bioaccumulated in shellfish and could pass to humans eating them<br>**1** It was assumed dilution would make it harmless (illustrates the need for the precautionary principle)<br>– now prohibited and replaced by copper<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely | 3 |
| **05.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** In the low-oxygen conditions of the spoil heap iron is in its soluble reduced form and is leached into the river<br>**1** In the river it is oxidised to insoluble iron oxide (orange sediment)<br>– the oxidation uses dissolved oxygen so the river is deoxygenated, killing aerobic organisms<br>**1 Control:** drainage water is passed over mesh screens where the iron oxidises and is deposited before the water reaches the river<br>– the solid iron is removed periodically<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leached** → nutrients or ions washed down out of the soil / washed away by water | 3 |
| **05.4** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** Lead-acid car batteries (recycled)<br>**1** Lead flashing on roofs / windows<br>**1** Radiation shielding<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |

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| Question | Marking guidance | Marks |
|---|---|---|
| **05.5** | 1 mark for the point below.<br>**1** They inhibit enzyme function, especially the enzymes of the nervous system / nerve cells<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11. | |
| **06.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 C:** Organic (methyl) mercury<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 1 |
| **06.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Anaerobic microbes converted the inorganic mercury into organic methyl mercury<br>**1** Methyl mercury is liposoluble so it is absorbed, crosses into the brain and biomagnifies along the food chain to fish eaten by people<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Most heavy metals are less soluble at high pH<br>**1** So they remain immobile and are not leached into water or absorbed by organisms<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **leached** → nutrients or ions washed down out of the soil / washed away by water<br>**Do not credit**<br>• 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-279; AQA spec 3.4.3.2.11. | |

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